

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

March-April 2018

CL 301 Transportation Engineering I

Date: 27.03.2018, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a)** Write short note on IRC. [05]
- (b)** Define Alignment. Enlist the ideal requirements of the alignment. [02]
- Q - 2 (a)** Mention the scales to which following drawing are usually prepared for Highway Project. [02]
- (1) Preliminary Survey Plans
 - (2) Detailed Cross-Section Plans
 - (3) Index Map
 - (4) Drawings of Cross Drainage and other Retaining Structures
- (b)** Explain superelevation provided on horizontal curve on highway with neat sketch [06]
- (c)** Write short note on stopping sight distance. [06]

OR

- (c)** Draw a neat sketch of Cross-Section of Road in embankment showing all its components. [06]
- Q - 3 (a)** Enlist methods of Surface Drainage of Highway. Explain Side Drain for Road in Embankment method with neat sketch. [05]
- (b)** Enlist the failures of flexible pavements. Explain Shear Failure Cracking with neat sketch. [04]
- (c)** What is Interface Treatment in Bituminous Construction? Explain Prime Coat used for interface treatment in bituminous construction. [05]

OR

- (c)** Write down construction procedure of Surface Dressing used for renewable of existing bituminous pavement. [05]

SECTION-II

- Q - 4** Explain different engineering properties of aggregate. Enlist different test performed [07]

on aggregate.

- Q - 5 (a)** Write Short Note on Rate of return method of the Highway Economics. [04]
(b) Why traffic surveys are carried out on highways? [05]
(c) Explain Group Index method to calculate the depth of different layers of flexible pavement. [05]

OR

- (c)** Explain the joints provided in rigid pavement. [05]
Q - 6 (a) Compare flexible and rigid pavements. [07]
(b) Explain Traffic Characteristics. [07]

OR

- (b)** Write short note on level of service. [07]
